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Detecting, mapping, and suppressing the spread of a decade-long *Pseudomonas aeruginosa* nosocomial outbreak with genomics

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eLife Assessment

This **important** work presents an example of how genomic data can be used to improve understanding of an ongoing, long-term bacterial outbreak in a hospital with an application to multi-drug resistant Pseudomonas aeruginosa, and will be of interest to researchers concerned with the spread of drug-resistant bacteria in hospital settings. The **convincing** genomic analyses highlight the value of routine surveillance of patients and environmental sampling and show how such data can help in dating the origin of the outbreak and in characterising the epidemic lineages. These findings highlight the importance of understanding environmental factors contributing to the transmission of P. aeruginosa for guiding and tailoring infection control efforts.

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Abstract

Whole-genome sequencing is revolutionizing bacterial outbreak investigation but its application to the clinic remains limited. In 2020, prospective and retrospective surveillance detected a *Pseudomonas aeruginosa* outbreak with 253 isolates collected from 82 patients in 26 wards of a hospital. Its origin was dated to the late 90s, just after the facility opened, and patient-to-patient and environment-to-patient cases of transmission were inferred. Over time, two epidemic subclones evolved in separate hosts and hospital areas, including newly opened wards, and hospital-wide sampling confirmed reservoirs persisted in the plumbing.

Pathoadaptive mutations in genes associated with virulence, cell wall biogenesis, and antibiotic resistance were identified. While the latter correlated with the acquisition of phenotypic resistances to 1st (cephalosporin), 2nd (carbapenems) and 3rd (colistin) lines of treatment, maximum parsimony suggested that a truncation in a lipopolysaccharide component coincided with the emergence of a subclone prevalent in long-term infections. Since initial identification, extensive infection control efforts guided by routine, near real-time surveillance have proved successful at slowing transmission.

Introduction

Pseudomonas aeruginosa is a versatile, opportunistic human pathogen causing a wide range of infections of both community and hospital origin[1–3]. In the clinic, patients with cystic fibrosis (CF), severe burns or neutropenia are of highest risk and cases are associated with high morbidity and mortality[3]. Furthermore, the management of these infections is increasingly difficult due to the global emergence of multidrug resistant (MDR) strains, especially carbapenem-resistant *P. aeruginosa* (CRPA), which now account for 16-30% of cases in the US[4]. In response, global and national agencies have categorized CRPA as a “Critical” pathogen posing a serious threat on public health[4].

While high-risk MDR lineages, exemplified by the most globally prevalent sequence type (ST) 235, are enriched in isolates carrying acquired carbapenemases and/or extended-spectrum beta lactamases (ESBLs)[5,6], the hallmark of *P. aeruginosa* lies in its remarkable propensity for developing resistances through chromosomal mutations[7]. Within the hundreds of intrinsic genes comprising this mutational resistome, several harbor well-characterized modifications that are the most common cause of resistance for nearly all classes of drugs[7]. These include gain-of-function mutations in gyrase subunits GyrAB (fluoroquinolones), penicillin-binding protein PBP3 (cephalosporins), various Mex efflux systems (multiple antimicrobials) as well as inactivation of the porin OprD (carbapenems) or the sensor component of the PhoPQ system, the latter associated with resistance to last-line polymyxin antibiotics[8–10].

Being capable of both acute and chronic infection, *P. aeruginosa* possesses a large arsenal of virulence factors such as a type 4 pili (T4P), O-antigens and lipopolysaccharides (LPS), extracellular enzymes, exotoxins (*e.g.* ExoS/U), a flagellum and five types of secretion systems (TSSs)[11]. These are generally part of complex regulatory pathways in which two-component systems (TCSs) often play a major role[11,12]. The ability of *P. aeruginosa* to form biofilm is a key aspect of its persistence in patients with chronic infections as well as in environmental niches[13]. In the hospital, biofilms are often found in water-related environments and eradication is particularly difficult due to increased disinfectant resistance[1,14]. This in turn poses a challenge for infection prevention and control (IPAC) as illustrated by the increasing reports of *P. aeruginosa* outbreaks and hospital-acquired infections (HAI) linked to contaminated sinks, drains or taps[15–18].

Here, routine genome-based surveillance of MDR clinical isolates across the US Military Health System (MHS) identified an epidemic cluster of *P. aeruginosa* in a single hospital in 2020. Retrospective analysis of isolates from the preceding ten years, together with Bayesian phylogenetics, revealed a protracted outbreak clone with cases from all hospital floors and an origin dated to the late 1990s. Dissection of this unique dataset sheds new light on the evolution, persistence, and routes of transmission of *P. aeruginosa* in the clinic.

Results

P. aeruginosa ST-621 is endemic in a large US tertiary hospital

The Multidrug-Resistant Organism Repository and Surveillance Network (MRSN) performed whole genome sequencing (WGS) of 5,129 *P. aeruginosa* isolates received between 2011 and 2020 from 71 facilities around the globe[19]. While a total of 547 STs were represented, high-risk epidemic lineages ST-235/*exoU*+ (10%) and ST-244/*exoS*+ (5%) were the most prevalent and widespread with isolates recovered from 37 and 28 facilities, respectively (**Fig. S1A**). By contrast, isolates from lineage ST-621/*exoS*+ also represented 5% of the total STs, but originated from just 6 facilities, with 89% from a single US tertiary hospital (Facility A) and sharing high-level genetic relatedness (0-23 allelic differences) by core genome MLST (cgMLST) (**Fig. S1B**). Furthermore, of the 5 additional facilities from which ST-621 was detected, patient transfer from facility A was observed in three instances and isolates from the remaining two were genetically distinct strains (88-89 allelic differences) (**Fig. S1B**).

Retrospective analysis revealed that a total of 253 ST-621 isolates were collected from facility A (a 425-bed medical center and the only level 1 trauma center in the Military Health System) between May 2011 and December 2020. The majority were isolated from respiratory (42%) and urine (39%) samples, followed by wound (7%), surveillance (6%), tissue (3%), and blood cultures (2%) (**Table S1**).

A protracted outbreak with cases dispersed throughout floors and wards

Phylogenetic analysis confirmed that all ST-621 isolates from facility A belonged to the same outbreak clone with an average distance (all vs all pairwise comparison) of just 38 single nucleotide polymorphisms (SNPs) and an interquartile range of 19 (**Fig. 1A**, **Fig. S2** and **Table S2**). A total of 82 patients (average age of. 57.2 +/- 21.0 and male/female ratio of 1.7) were represented, with the majority (56%) contributing serial isolates. In particular, six patients (ID 10, 27, 34, 35, 49 and 63) showed prolonged carriage, best exemplified by patient 35 from whom 21 urine isolates were collected over 4.5 years (**Fig. 1A** and **B**).

Two outbreak subclones, designated SC1 and SC2, were identified from the tree topology and showed increased genetic relatedness within each (average distance of 24 and 31 SNPs, respectively) (**Fig. 1A**). Overall ST-621 isolates were observed in 26 wards/units from all 7 floors of two buildings (Main and Tower Buildings) in facility A (**Fig. 1C**), yet subclones SC1 and SC2 were spatially and temporally distinct. 78% of SC1 isolates were collected from patients in the Main Building and 60% were detected before 2014. In contrast, 52% of SC2 isolates originated from patients in the Tower Building and 77% were collected after 2014 (**Fig. 1A**). Specifically, the early spread of SC1 was largely connected to patients in adjacent ICUs and step-down units on the 2nd and 3rd floors (wards 1-10) of the Main Building. The subsequent emergence of SC2 largely centered in patients receiving care in adjacent ICUs (wards 20-21) on the 3rd floor within the Tower Building (**Fig. 1A** and **C**). Throughout the outbreak, new cases due to either SC1 or SC2 were also detected from patients in the emergency room (ER). Unlike the isolates from other locations, these ER isolates were nearly all (32 vs 92%, $p < .0001$) from urine cultures and, as expected, the majority (72%) were serial isolates from returning patients at Facility A (**Fig. 1C**, **Table S1**). The remaining were collected from 9 patients for whom a previous visit at Facility A is suspected but could not be established as no previous isolates were collected from them and no historical patient data was available.

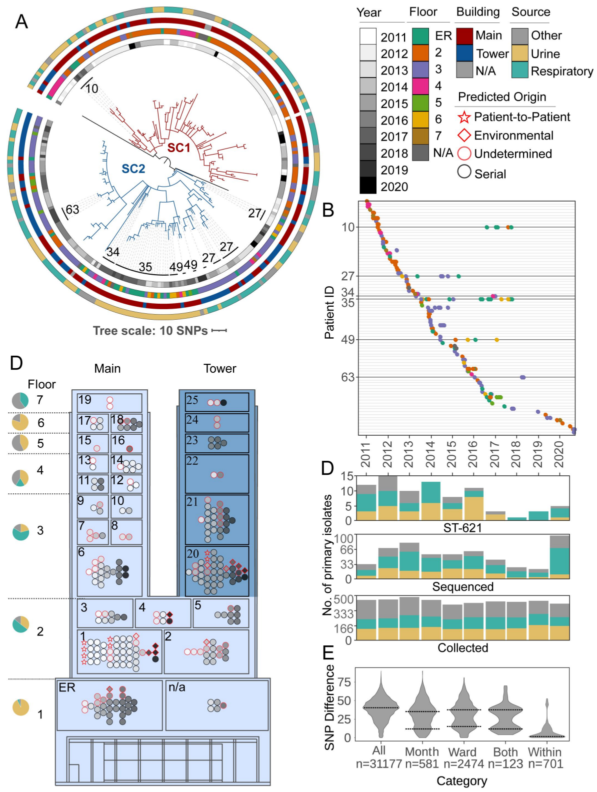

Figure 1.

Phylogenetic and spatio-temporal analysis of 253 clinical ST-621 *P. aeruginosa* isolates.

(A) SNP-based, core genome phylogeny. Outbreak subclones are colored with red (SC1) or blue (SC2) branches. From inner- to outermost, datasets show: the year, the floor, the building, and the source of isolation. Patients with prolonged carriage (>1 year) are indicated. **(B)** Timeline of isolate collection. Each line indicates a single patient. Isolates are shown as dots colored by the floor the patient was on. Patients with prolonged carriage are highlighted. **(C)** Simplified floor plan of the Main (light blue) and Tower (dark blue) Buildings of Facility A. Wards and floors are numbered. For each floor, a pie chart indicates the proportion of isolates collected from urine, respiratory, or other cultures. Within each ward, isolates are colored by year of collection with a red outline indicating primary (first) isolates. Symbols indicate the presumed origin of infection. **(D)** Bar charts showing for each year the number of: all *P. aeruginosa* primary clinical isolates collected at Facility A (bottom), the subset of genome-sequenced primary isolates (middle), and the subset of identified ST-621 isolates (top). Bars are color-coded by the isolation source. **(E)** Distribution of SNP distances for all inter-patient isolates or subsets collected in the same month, the same ward, or both. SNP distances within patient isolates are also indicated. Dotted lines indicate the local maxima.

The epidemic curve revealed that between 2011 and 2016 an average of 11.5 new cases were detected every year. From 2017 to 2021, only 12 new patients were identified (3 cases a year) (**Fig. 1D**). Although the rate of outbreak cases appeared to decrease, the overall number of *P. aeruginosa* cases in facility A (irrespective of lineages) remained stable throughout the studied period. However, due to staffing issues, the number of isolates received/sequenced by the MRSN every year was not consistent (**Fig. 1D** and **Fig. S3**).

An origin dated soon after the opening of facility A

Though the collection of clinical isolates at facility A only started in 2011, genetic diversity was already evident (16.3 SNPs on average), suggesting an earlier origin (**Table S2**). Molecular clock phylogenetic analyses of all studied isolates inferred the time of the most recent common ancestor (MRCA) to be in 1999 (± 3 years), soon after the opening of the Main Building of facility A in 1996 (**Fig. 2**). Furthermore, this analysis predicted that, despite the observed temporal succession in sequenced isolates, SC2 was not a descendant of SC1. Instead, the two subclones emerged concurrently with respective MRCAs dated in 2004 (± 2 years) (**Fig. 2**).

Despite this predicted early origin, it is only after facility expansion and the opening of the Tower Building in 2012 that the first SC2 isolates were ever sampled (**Table S1**). Specifically, the earliest SC2 isolate collected was from a wound infection in patient 27. Remarkably, the 15 serial isolates collected over 3 years (2012 to 2015) from this patient were not monophyletic and their MRCA coincided with the emergence of this subclone (**Fig. 2**).

Environmental reservoirs sustain nosocomial transmission

The presumed origin of an infection, either directly from patient-to-patient or from environment-to-patient, was designated based on SNP thresholds (starting from pairwise comparisons for all inter-patient isolates, **Table S2**) and known epidemiological links, or lack thereof. Indeed, while a unimodal distribution of SNPs (peak at 40.5) was observed for the whole cohort, a bimodal profile appeared when inter-patient isolates from the same month, the same ward, or both were selected (local maxima of the lower peak at 12.4, 15.3, and 11.6 SNPs respectively) (**Fig. 1E**). This pool of virtually identical isolates from patients with spatial and temporal overlap represented the most likely cases of direct, patient-to-patient transmission. Pairs of isolates from only 10 patients satisfied these conditions (**Table S1**): All carried subclone SC1, occurred in 2011 and 2012 (with two exceptions), and six were hospitalized in ICU #1 in the Main Building of facility A (**Fig. 1C**).

In contrast, a presumed environmental origin was designated when a new patient was infected with a virtually identical strain from a preceding patient in the same ward, but no known temporal overlap was established for at least 1 year (based on isolates collection dates). Twelve cases met these criteria (**Fig. 1C** and **Table S2**). Among them, 3 patients (ID 57, 61, and 74) in ICU #20 on the Tower Building had primary isolates (first isolates from a patient) of subclone SC2 that were distinct by only 0-4 SNPs despite being collected years apart (in 2015, 2016, and 2018, respectively) (**Fig. 1C**). Furthermore, an origin from contaminated surfaces was also predicted for 7 of the 9 most recent patients (2018-2020). This included 5 cases caused by the apparent resurgence of SC1 in ICUs #1-2 of the Main Building (**Fig. 1C**).

Genomic inferences were supported by contemporary environmental sampling, conducted in 2021 and 2022, during which a total of 159 swabs were taken from 55 distinct locations in 8 wards/units including the ER in Facility A. This confirmed the existence of reservoirs (*i.e.* sink drains located in patient rooms) of both SC1 and SC2 isolates (n = 17) in 7 distinct patient rooms in wards #1, 2, 6 and 20 throughout the Main and Tower Buildings of facility A (**Fig. 3**, **Table S3**). In that timeframe, an additional 56 clinical isolates (25 SC1 and 31 SC2) from 13 patients (including 11 new cases) were collected, indicating transmission events were still occurring (**Fig. 3**).

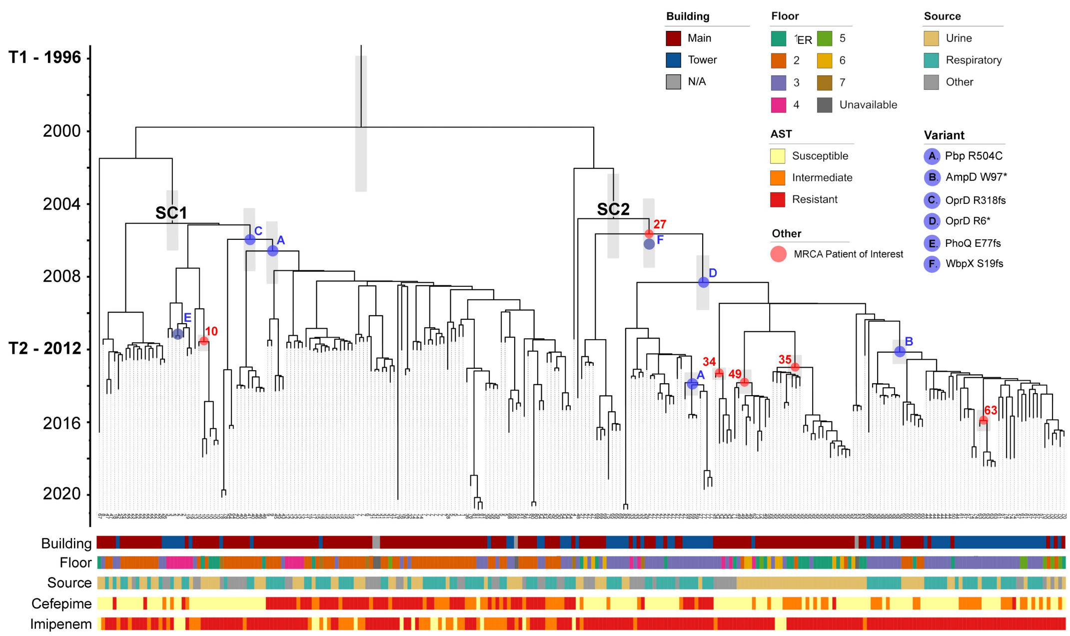

Figure 2.

Dated phylogeny of 253 ST-621 outbreak isolates.

Time-stamped phylogeny. Error bars for annotated nodes are shown as gray boxes. Years are indicated on the left, along with the opening of the Main Building (T1), and the Tower Building extension (T2). The MRCA of SC1 and SC2 are indicated in black text above the node. The MRCA of isolates from patients with prolonged carriage are shown in red, indicated by their patient ID. The predicted emergence of select variants are shown in blue. Data showing the floor and source isolates were collected from is shown, as in **Figure 1**. Select datasets are shown including the antibiotic susceptibility testing results for cefepime (cephalosporin) and imipenem (carbapenem).

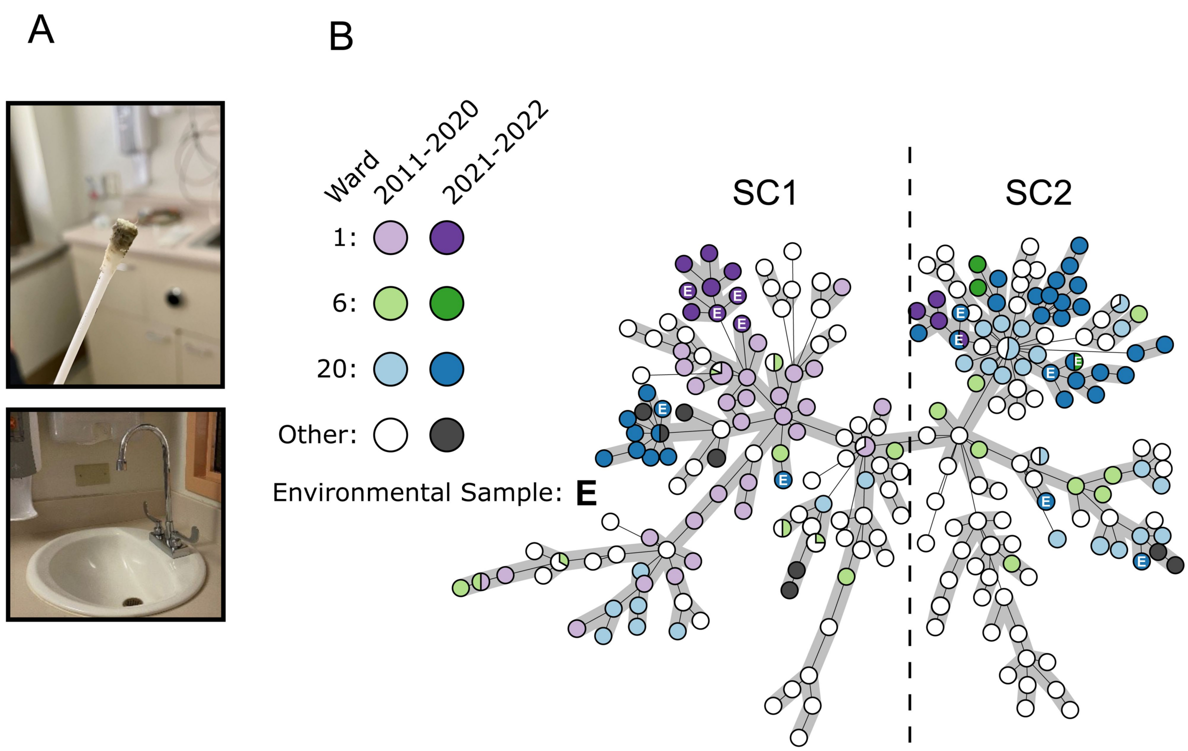

Figure 3.

Identification of environmental reservoirs of the ST-621 outbreak clone in sink drains.

(A) Photos depict a swab from a sink drain in a patient room and the typical sink design (shallow basins and gooseneck faucets) in facility A. **(B)** Minimum Spanning Tree of 326 Facility A ST-621 *P. aeruginosa* isolates, including contemporary isolates collected in 2021 and 2022 from clinical (n= 56) and environmental (n= 17) sources. Isolates from wards #1, 6 and 20 where environmental contamination was identified are colored. Environmental isolates are indicated with the letter E.

Selection on resistance genes and rising prevalence of CRPA due to porin defects

Phenotypically, a prototypical outbreak isolate showed non-susceptibility to aminoglycosides (except for amikacin), 3rd and 4th generation cephalosporins, carbapenems, and fluoroquinolones (**Table S4**). However, resistance profiles were not homogenous and temporal patterns were observed (**Fig. 4A**). In particular, while 63% of the isolates collected in 2011 were non-susceptible to imipenem, this fraction gradually increased to >95% by 2014 and beyond. In the same timeframe, the prevalence of isolates with non-susceptibility to cefepime decreased from 81% to 37% (**Fig. 4A**). Both phenomena were linked to the decline of SC1 and rise of SC2, with the latter displaying significantly higher (98% to 76% in SC1, chi-square = 53.5 and $p<0.001$) and lower (38% to 66%, chi-square = 25.3 and $p<0.001$) rates of resistance to imipenem and cefepime, respectively (**Fig. 2**). In contrast, resistance to tobramycin and ciprofloxacin was highly prevalent and remained relatively stable over the sampling period (**Fig. 4A**).

Genotypically, outbreak isolates contained no plasmids (based on the finished genomes of a diverse subset of isolates, **Table S1**) and, besides an intrinsic AmpC-type β-lactamase, only carried two acquired resistance genes, known to confer resistance to ciprofloxacin (*crpP*) and tobramycin (*aac(6')-Ib4*) (**Table S1**). To explain other phenotypic resistances, and in the absence of an acquired ESBL or carbapenemase, a genome-wide analysis of non-synonymous (NSY) and loss of function (LOF) mutations was performed for all outbreak isolates (**Table S5**). From a list of 164 genes (**Table S6**) for which known chromosomal mutations have been linked to antibiotic resistance[11–13], 68 distinct mutations (including 20 predicted LOF) in 37/164 genes were identified in at least one outbreak isolate (**Table S7**). Of particular interest, 8 genes (or functional clusters) were each mutated independently three or more times (**Fig. 4B**).

The most frequently mutated gene was *oprD*, encoding a porin that facilitates the diffusion of amino acids, small peptides, and carbapenem antibiotics into the cell[8]. Five of the six independent mutations identified in OprD were predicted LOF, including an R6* stop codon and a R318fs frameshift observed in 115 SC2 and 92 SC1 isolates, respectively (**Fig. 4B**, **Table S7**). Phylogenetic Analysis Using Parsimony (PAUP) predicted that both the R318fs and R6* mutations occurred early (between 2006-2008) within the evolution of their respective subclones (**Fig. 2**). Despite this early origin, highly carbapenem resistant SC2-OprD-R6* isolates were first sampled from patients in mid-2012 and only became predominant from 2014 onward (**Fig. 2**).

Within all other mutations in antibiotic resistance associated genes (**Table S7**), only two more variants were found in > 50 outbreak isolates: a W97* stop codon in β-lactamase expression regulator AmpD and a R504C substitution in penicillin-binding protein PBP3 (**Fig. 4B**). The LOF in AmpD, predicted to have emerged in 2012, was observed in a subset of 51 monophyletic SC2-OprD-R6* background isolates that were not consistently resistant to cefepime (**Fig. 2**). In contrast, the R504C substitution in PBP3 was acquired independently twice (within SC1-OprD-R318fs in 2007 and within SC2-OprD-R6* in 2014) and was strongly associated with a resistance to 4th-generation cephalosporins (**Fig. 2**). Overall, *ampD* and *pbp3* were among the genes with the most independently acquired NSY or LOF mutations, with six and four distinct sites, respectively (**Fig. 4B**). Accumulation of mutations, albeit in small numbers of isolates, was also observed in other genes involved in β-lactam resistance (*e.g.* catabolite repression control *crc*, and *mpl* UDP-muramic acid/peptide ligase) as well as antibiotic efflux systems (*mexXY* and *mexCD-oprJ*) (**Fig. 4B**).

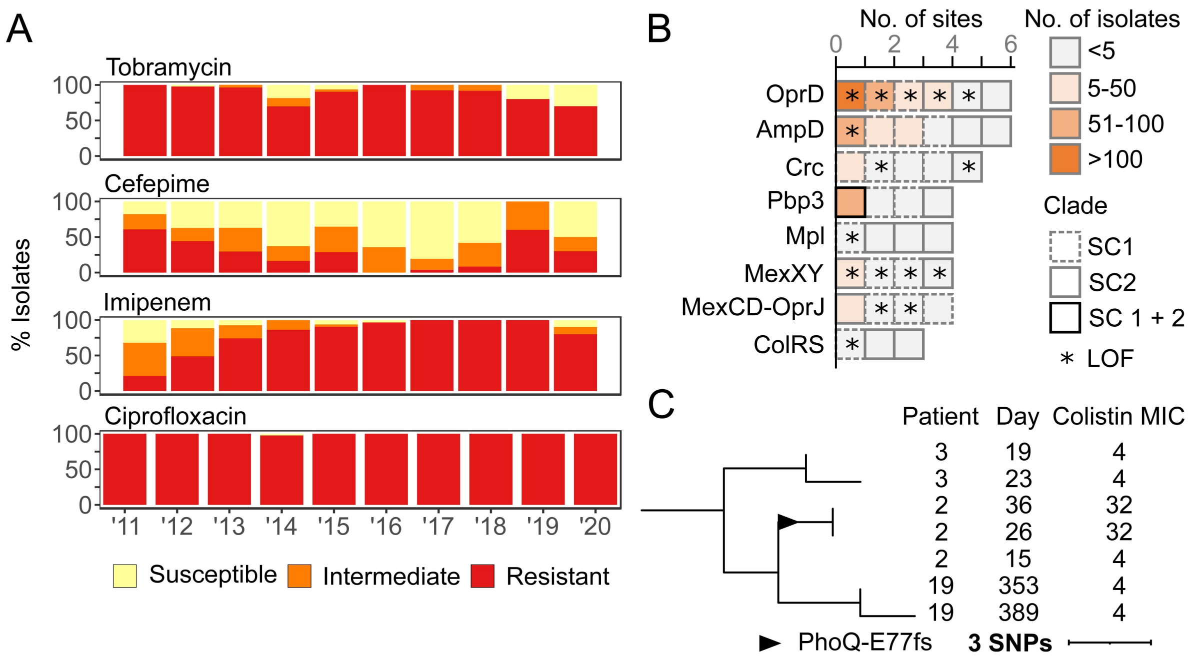

Figure 4.

Antibiotic resistance and poly-variant sites of 253 outbreak isolates.

(A) Bar charts showing the proportion of clinical isolates resistant to tobramycin, cefepime, imipenem, and ciprofloxacin each year from 2011-2020. **(B)** Chart of mutations (excluding synonymous) in resistance-associated genes. Each block indicates one distinct mutation observed within the corresponding gene. Blocks are colored by the number of isolates sharing this exact mutation. Blocks are outlined to indicate whether the mutation is found within SC1, SC2, or isolates from both subclones. Mutations causing a predicted loss-of-function (LOF) are indicated with a star. **(C)** Pruned SNP-based phylogeny showing the acquisition of colistin resistance in Patient 2. Patient ID, day of collection (since the first outbreak isolate), and available colistin MIC are shown on the right. The MRCA of the PhoQ E77fs mutation is indicated with a triangle.

Sporadic emergence of colistin resistance

Though rare, the presence and impact of mutations in genes associated with resistance to colistin, a last-line antibiotic, was examined. NSY mutations were detected in the TCSs ColRS (three distinct sites in four isolates) (**Fig. 4B**) and PmrAB (two distinct sites in three isolates) (**Table S7**) but colistin MIC's were the same as wild-type isolates (4 mg/L). By contrast, patient 2 (who had a history of colistin therapy) acquired a predicted LOF mutation (E77fs) in sensor protein PhoQ in two serial isolates resulting in an 8-fold increase in MIC (32 mg/L) (**Fig. 4C**). Notably, the emergence of this colistin resistant strain, which occurred in early 2011 (**Fig. 2**), remained constrained to a single patient and no further spread was detected.

Mutational convergence in virulence, cell wall biogenesis and signaling pathways

Genome-wide, and excluding synonymous mutations, 64 genes were each mutated independently two or more times over the course of this outbreak (**Table S5**). Compared to the distribution observed for the whole genome, these 64 genes were functionally enriched for signal transduction (28% vs 6% in the whole genome, chi-square $p<0.001$) and cell wall/membrane biogenesis (17% vs 7%, $p= 0.0017$) (**Fig. 5A**). For the latter category, besides genes already discussed for their direct role in antibiotic resistance (i.e *pbp3, oprD, mpl, and mex* loci*)*, six other genes were recurrently mutated in outbreak isolates including LPS biosynthesis *rmlA* and choline transporter *betT2* (**Fig. 5A**).

For genes associated with signaling functions, various TCS involved in pathogenesis were represented, including *narXL*, *fleRS*, and the catabolic regulation *cbrAB* (**Fig. 5A**). Notably, *cbrAB* was the most independently mutated loci in this dataset with 16 distinct NSY mutations in 80 isolates from both subclones. Beyond TCS, sigma factor *pvdS*, required for toxin expression (6 NSY and 1 LOF), motility regulator *morA* (11 NSY and 2 LOF) and mucoidy regulatory system *mucAB* (4 sites including three predicted LOF) were also recurrently mutated (though rarely fixed) in outbreak isolates (**Fig. 5A**).

Finally, at the level of functional pathways, and compared to the theoretical frequency for an even distribution of mutated sites across the whole genome (0.11 sites per kb of coding sequence), NSY and LOF mutations were significantly enriched in genes involved in the biosynthesis of the Type IV secretion system (0.59 sites/kb, chi-square $p<0.001$), flagellar biosynthesis and chemotaxis (0.21 sites/kb, $p= 0.0264$), alginate production (0.31 sites/kb, $p= 0.0088$) and O-antigen production (0.36 sites/kb, $p=0.0126$) (**Fig. 5B**, **Table S8**). Importantly, the WbpX D-rhamnosyltransferase, part of the O-antigen biosynthesis pathway, was disrupted (S19fs) in 127 SC2 isolates (**Table S5**). PAUP predicted this frameshift occurred in 2005 (± 24 months), concomitant with the predicted emergence of SC2 (**Fig. 2**).

Discussion

WGS technology is revolutionizing infection prevention and control. Here, routine surveillance of MDR pathogens across the US Military Health System was critical in uncovering a decades-long *P. aeruginosa* epidemic cluster. With traditional approaches alone[20], comparable outbreaks may avoid detection due to the sporadic nature of patient infections, scattering of cases throughout the hospital, and changing antibiotic susceptibility profiles. Furthermore, because of budget constraints, surveillance programs focusing on "high-risk" isolates carrying select resistance genes (*e.g.* ESBLs, carbapenemases)[4–6], would also fail to detect this ST-621 outbreak clone. Indeed, while the ST-621 lineage was the focus of several studies in the late 2000's, these examined a *bla*$_{IMP-13}$ carrying epidemic clone which spread throughout Europe and has since been

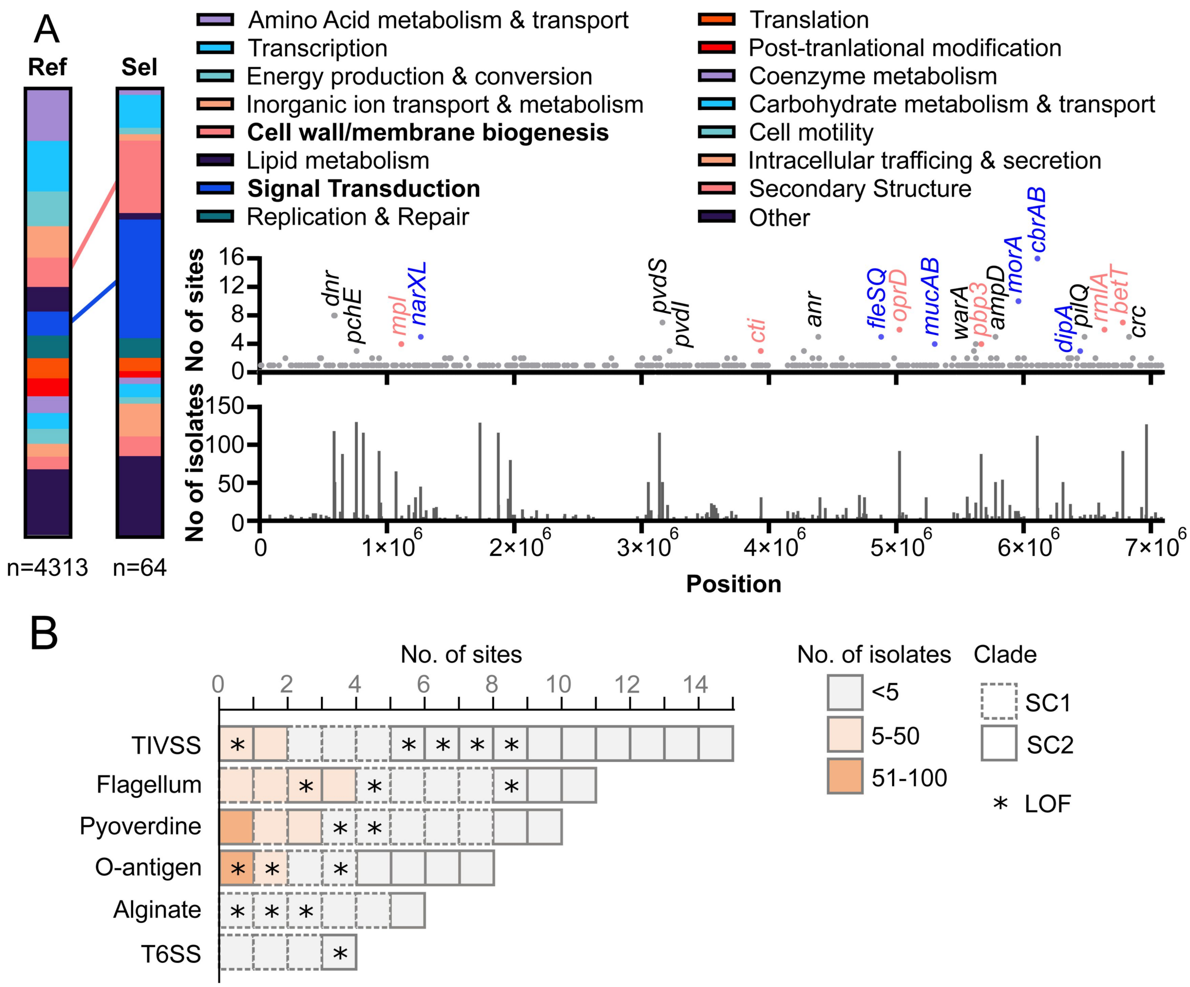

Figure 5.

Identification of pathoadaptive mutations.

(A) Distribution of COG functional categories for all genes (Ref) compared to the subset of genes (Sel) with 2 or more distinct mutated sites (excluding synonymous) in outbreak isolates. Significant enrichment in cell wall biogenesis and signal transduction categories are indicated. A positional chart of mutations across the entire chromosome of closed reference genome for isolate 4605 is provided. The number of unique mutation sites within each gene is indicated (top) as well as the number of outbreak isolates which carried each mutation (bottom). Genes with 3 or more distinct mutated sites are labeled and genes involved in cell wall biogenesis and signal transduction genes are color coded. **(B)** Accumulation of mutations (excluding synonymous) in genes and pathways associated with *P. aeruginosa* pathogenesis. Each block indicates one distinct mutation observed within a gene part of the indicated functional group (data available in Table S8). Blocks are colored by the number of isolates sharing this exact mutation. Blocks are outlined to indicate whether the mutation is found within SC1, SC2, or isolates from both subclones. Mutations causing a predicted loss-of-function (LOF) are indicated with a star.

sporadically detected in South America and Southeast Asia[21–24]. To our knowledge this IMP-carrying strain has not been reported in the USA to date, and the outbreak in Facility A is the result of a distinct MDR clone lacking the carbapenemase.

Similar to the US Military Health System, healthcare networks worldwide that already benefit from prospective WGS surveillance programs are reporting large numbers of unrecognized outbreaks[18]. However, two resources rarely available in other settings were key to this investigation: a genome repository of isolates from the preceding decade and the ability to conduct prospective environmental sampling. The former was integral to date the presumed origin of the outbreak to the late 90s, soon after the hospital opened, and to trace the evolution and separate spread of two subclones throughout floors and wards. The latter revealed that, more than 20 years later, reservoirs of both subclones persist in sink drains from patient rooms throughout the facility. This supports previous evidence implicating these sites as major reservoirs for *P. aeruginosa* in the clinic[1,25]. Decontamination of drains can be extremely challenging due to the limited penetration of disinfectants, the lack of access (*e.g.* to perform scrubbing), recolonization (*e.g.* disposal of contaminated patient specimens) or retrograde growth from p-traps[1,14,25]. Furthermore, sinks with shallow basins and gooseneck faucets directing water straight into the drain, similar to those in Facility A (**Fig. 3A**), have been linked to increased back splash onto nearby surfaces and medical equipment[26].

Besides environmental contamination, reservoirs within patients likely contributed to the spread and longevity of the outbreak. In particular, long-term infections, documented for a significant fraction of patients, provided a recurring source of the epidemic clone. It is noteworthy that 5 of the 6 patients with long-term (> 1 year) infections (and 9 out of 11 if reduced to 6 months) were carriers of subclone SC2. Compared to SC1, all but one SC2 isolates carried a truncated WbpX glycosyltransferase, an essential component of the common polysaccharide antigen (CPA) biosynthesis[27]. While this mutation could result in increased immune evasion, the rough LPS phenotype recurrently observed *in vivo* (in particular in CF patients) usually results from a functional CPA but the lack of O-specific antigen[11,28]. In further evidence for a progression toward a host-adapted lifestyle, a convergence of mutations in genes involved in alginate production, quorum-sensing deficiency, loss of motility, and decreased protease secretion, all phenotypes associated with chronic infections[29], was observed throughout the 10 years of sampling.

Considering its role in long-term infections and the distinctive association with patients and sinks in the Tower Building of Facility A, the origin and emergence of subclone SC2 is of particular importance. Although obfuscated by the limited sampling, a plausible scenario (supported by BEAST2 inferences) would be that a patient with an initial infection acquired during a stay in the Main Building before the SC1-2 split (ca. 2004) proceeded to shed the epidemic strain in the newly opened (2012) Tower Building during successive visits. Importantly, the Main and Tower Buildings of facility A have distinct plumbing systems, providing the necessary ecological separation (minus inter-hospital patient transfers) for the further spread and divergence of the two sub-clones. Patient 27 is just such an example, but the lack of early sampling (2000-2010) and incomplete collection thereafter precludes a definitive answer. Indeed, other scenarios cannot be discounted, with a recent study showing sinks in a newly built ICU were already contaminated with an outbreak clone of *P. aeruginosa* before the arrival of the first patients[17].

One of the most notable features of this outbreak was the evolution of antibiotic resistance over time, with the emergence of resistance to 1st (cephalosporin), 2nd (carbapenems) and 3rd (colistin) line treatments. The potent R504C substitution in PBP3 was selected in each subclone, and in addition to facilitating cephalosporin resistance, has also been linked to ceftazidime/avibactam resistance[30]. While no patient prescription data is available from Facility A, it can be speculated that the high prevalence of cephalosporin resistance in early outbreak isolates led to increased carbapenem use, which in turn selected for the many independently evolved OprD mutants, one of the most common mechanisms for carbapenem resistance in *P. aeruginosa*[11]. Finally, albeit limited to a single patient, the emergence of colistin resistance, through a well characterized mechanism, is a reminder that the threat of extensively drug resistant *P. aeruginosa* is only a prescription and a few mutations away[31].

Some limitations were noted for this study and complicated the genomic inferences. First, although criteria remained the same, the partial and inconsistent sampling of clinical *P. aeruginosa* isolates prevented from capturing the exact magnitude of the outbreak through the years and sampling biased cannot be ruled out. Second, detailed epidemiological data for the patients (date of admission(s), bed location, reason for sampling, medications prescribed, etc.) were unavailable. As a result, possible patient overlaps were only inferred from the location and culture date of the isolates, and many were likely not detected. Third, fixed thresholds (*i.e.* genetic relatedness, time between isolates collection, and shared hospital location) were applied to predict a possible origin of infection. Acknowledging the other limitations, these were designed to predict the most likely cases of direct patient-to-patient transmission, when an overlap was identified, or environment-to-patient transmission, when a patient overlap was ruled out. Because of conservative criteria, the origins of most cases remained undetermined and an alternative to fixed threshold, using TransPhylo[32], is being explored in a follow-up study. This approach uses a probabilistic model to predict whom infected whom in outbreak scenarios and reconstruct transmission trees[32]. Fitting our dataset, TransPhylo was recently extended to allow the use of multiple genomes per host[33] and to incorporate epidemiological data into the analysis[34].

The data generated during this study has resulted in various ongoing interventions (*e.g.* closing sinks, replacing tubing, using foaming detergents[35]) at Facility A. Sampling and sequencing, in near real-time, of all *P. aeruginosa* clinical isolates (and not just MDR) also commenced in 2021. Notably, as of May 2025, no new ST-621 patient cases have been reported in over 15 months, and the unbiased contemporary data using all *P. aeruginosa* from patients at this facility confirms the spread has slowed, with just 4 cases identified in 2022-2024. Similar to Facility A, the roll-out of routine WGS surveillance in the clinic has the potential to improve patient care and prevent some of the estimated 136 million hospital-associated drug-resistant infections per year globally[36].

Materials and methods

Data Collection & Antibiotic Susceptibility Testing

As mandated by the US National Action Plan for Combating Antibiotic-Resistant Bacteria (CARB), the Multidrug Resistant Organism Repository and Surveillance Network (MRSN) collects and analyses clinically relevant MDR organisms across the U.S. Military Health System (MHS) and around the world in collaboration with the Global Emerging Infections Surveillance (GEIS) Branch. All isolates received by the MRSN underwent whole genome sequencing (single colony from a pure culture) and criteria for isolate collection were as follow. Between 2011-2020, all MDR isolates (resistance to ≥ 3 antibiotic families) were requested from all US facilities, but compliance varied between years and facilities. For Facility A, starting in 2021, all clinical *P. aeruginosa* isolates (MDR or not) were requested.

As a result of this sampling, from 2011 to 2020, 5,129 MDR *P. aeruginosa* isolates were collected from 71 facilities and sent to the MRSN. Of these, 1,511 were collected from Facility A, a large tertiary U.S. military hospital located in the continental United States. In total, 253 isolates belonged to the ST-621 outbreak clone.

Antibiotic susceptibility testing (AST) was performed in a College of American Pathologists (CAP)-accredited lab at Facility A using a Vitek 2 (card GN AST 71 and GN ID; bioMérieux, NC, USA). For isolates without MICs, AST was performed in the MRSN CAP-accredited clinical lab as previously described[37].

Whole Genome Sequencing & Bioinformatic Analysis

Genomic DNA was extracted and sequenced via Illumina MiSeq or NextSeq benchtop sequencer (Illumina, Inc., San Diego, CA) as previously described[37]. Long read sequencing was performed using PacBio RS II (Pacific Biosciences of California, Inc., Menlo Park, CA). Kraken 2 was used to identify isolate species and check for contamination[38]. Short-read sequencing data were trimmed for adapter sequence content and quality using bbduk[39]. *De novo* assembly was performed using Newbler v2.9[40]. Minimum thresholds for contig size and coverage were set at 200 bp and 49.5×, respectively. Long-read sequencing data were assembled using HGAP 3.0 in the SMRT Analysis portal. *In silico* multilocus sequence typing (MLST) was performed using the scheme developed by Curran and Dawson[41]. Antimicrobial resistance genes were annotated using a combination of AMRFinderPlus and ARIBA[42,43]. The genomes of all 253 ST-621 isolates have been deposited in the National Center for Biotechnology Information under BioProject PRJNA852179.

Core Genome MLST, Gene Annotation, SNP Calling, and Phylogenetic Analysis

Core genome MLST were determined in SeqSphere + (Ridom, Germany) using the cgMLST scheme developed by Curran *et al*[41] and a 90% cut-off (3,960 of 4,400 genes). SNP calling was performed with Snippy v.4.4.5 (*https://github.com/tseemann/snippy*) using error corrected [Pilon v1.23][44] and annotated [Prokka v1.14.6][45] draft assembly, with isolate MRSN 4605 as the reference (chosen because it was both collected early in the outbreak and had low root to tip divergence). Type strain PAO1 was used as the basis for annotation[46]. The core SNP alignment was filtered for recombination using Gubbins v2.3.1[47]. A SNP-based phylogeny was created by inferring a maximum-likelihood tree with RaxML-NG v0.9.0[48] using GTR+G (50 parsimony, 50 random). The phylogeny was visualized in iTOL[49].

Gene annotation and clustering showed 8,754 total genes in the pangenome of the outbreak. The core genome (present in ≥ 99% of isolates) was comprised of 5,674 genes, with a shell genome (defined as genes found in ≥ 95% of isolates) of 297 genes, and a cloud genome (≥ 15% of isolates) of 530 genes.

Curation and analysis of recurring mutations in outbreak isolates

Genome-wide analysis of non-synonymous and loss-of-function mutations in all outbreak isolates showed a total of 629 mutated sites (257 singletons and 372 sites shared by ≥ 2 isolates) in 488 distinct genes (**Table S5**). For variant sites identified in >99% of outbreak isolates, individual nucleotide-based sequence alignments were performed using a wild-type allele from the PAO1 genome (*https://www.pseudomonas.com/*). This allowed the identification of sites for which the MRSN 4605 reference genome was carrying the mutation and not the remaining outbreak isolates. These were curated (e.g. entry was modified to reflect the true mutation and its presence/absence in all outbreak isolates) manually and indicated as such in **Table S5**.

SNP distances between isolates from different patients

SNP distances were obtained from the alignment by using snp-dist (*https://github.com/treemann/snp-dists*). To create violin plots, the complete all-vs-all dataset was limited, based on 5 sets of criteria, to pairs of isolates that (1) originated from different patients; (2) originated from different patients, and were collected in the same month; (3) originated from different patients and were collected in the same ward; (4) originated from different patients, were collected in the same month, and the same ward; or (5) originated from within the same patient (**Fig. 1E**).

Primary isolates were designated as originating from a Patient-to-Patient transmission if they were part of a pair from different patients, were collected in the same ward, were separated by ≤ 10 SNPs (to account for the observed within-host variability, **Fig. 1E**), and were collected ≤ 31 days apart (as a proxy for a possible patient overlap). Primary isolates were conservatively designated as environmental if they were part of a pair from different patients, were collected in the same ward, were collected > 365 days apart (as a proxy for unlikely patient overlap), and were separated by a number of SNPs ≤ 3x the number of years they were separated to account for the 2.987E-7 substitutions/site/year from the BEAST analysis. This conservative threshold was chosen to favor high specificity as this inference relied solely on clinical isolates (i.e. the identical environmental strain in the patient-environment-patient chain was not sampled). For these clinical isolates to have acquired no/very little mutation in that much time, no/low replication is expected and, although unsampled, we propose this most likely happened on hospital surfaces.

In each of these cases, direct transmission or transmission mediated by a healthcare worker were not distinguished as no sampling was available from the later. All other primary isolates were designated as Undetermined.

Bayesian evolutionary phylogenetic analysis

To evaluate the strength of the temporal signal, TempEst v1.5.3 was utilized to visualize the relationship between root-to-tip genetic distances for samples with known collection dates[50]. To date internal nodes of interest on the phylogeny, bayesian phylogenetic inference was performed using BEAST2 v2.6.5 on a recombination free alignment, removing samples with uncertain collection dates, and accounting for constant sites with beast2_constsites (*https://github.com/andersgs/beast2_constsites*)[51]. The HKY substitution model was selected based on evaluation of all possible substitution models in bModelTest v1.2.1[52]. The random clock model was selected based on support by the marginal likelihood value using the Nested Sampling package v1.1.0[53]. BEAST2 was run under a coalescent constant population model with a Markov chain Monte Carlo length of 1 x 10^8 sampling every 5 x 10^3 steps. Analyses were repeated five times to confirm consistency between the obtained posterior distributions. Parameter estimates were computed using Tracer v1.7.1. Posterior trees were combined with LogCombiner and summarized in TreeAnnotator after a 50% burn-in. The final MCC target tree was visualized and annotated using iTOL (*https://itol.embl.de/*)[49].

Data and materials availability

Both genomic assemblies and raw sequencing data of all isolates analyzed in this study are publicly available in the NCBI database under the BioProject number PRJNA8852179

Figure supplements

Figure S1.

Minimum Spanning Trees of *Pseudomonas aeruginosa* cgMLST.

Isolates are color coded by facility, showing the 10 most prominent. **(A)** Includes all 5,129 *Pseudomonas aeruginosa* in the MRSN's collection from 2011-2020. Clusters of isolates belonging to ST-235, -244, and -621 are indicated. **(B)** Includes all ST-621 isolates from 2011-2020. Genetic distances higher than 23 allelic differences are indicated.

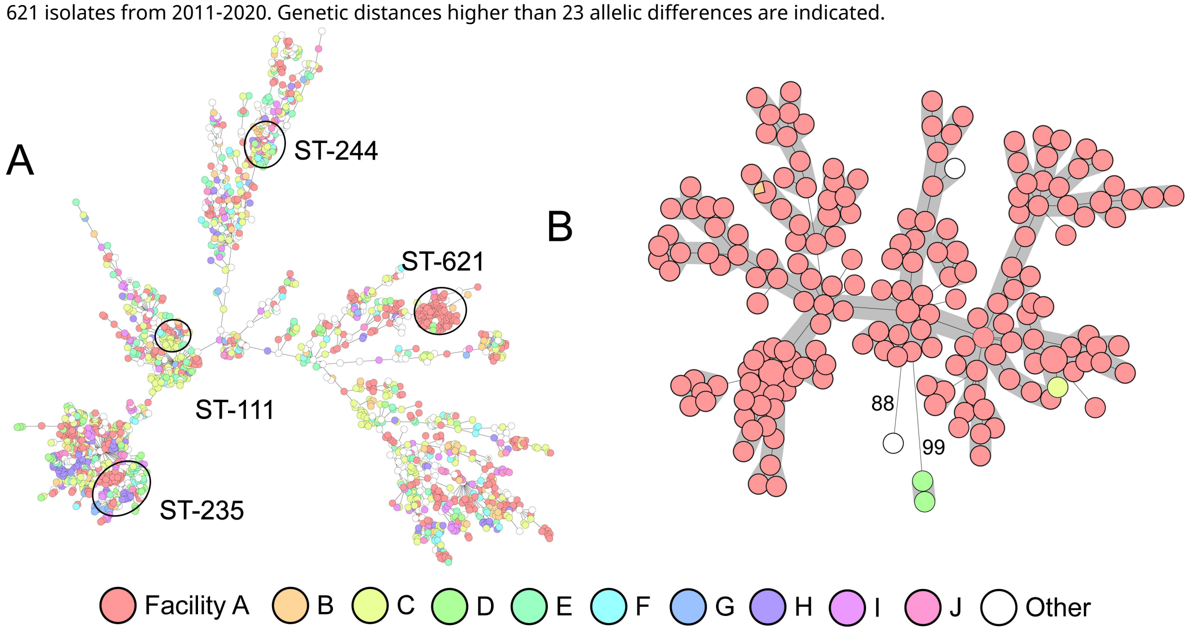

Figure S2.

Bar chart showing the distribution of SNP distances for all inter-patient isolates.

For a given SNP distance (x-axis) the number of observed isolate pairs is plotted (y-axis). Raw data available in Table S2.

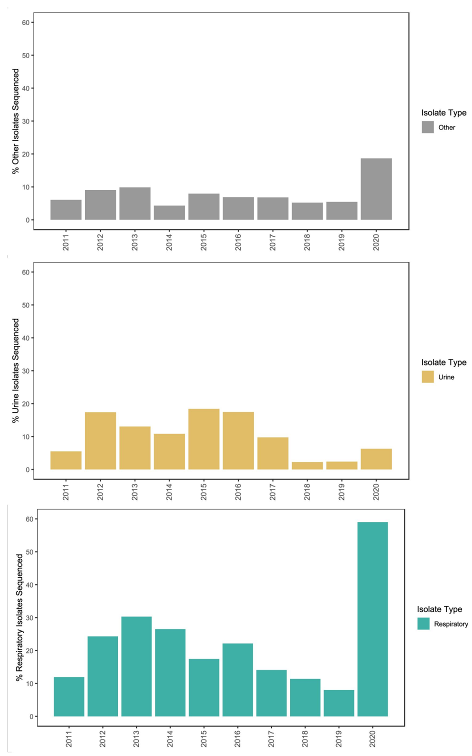

Figure S3.

Bar charts showing the percentage of sequenced *P. aeruginosa* isolates compared to the total collected by Hospital A stratified by year (2011-2020) and isolate types (respiratory, urine, or other).

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Additional information

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Author Contributions

P.T.M and F.L. designed research; W.S., L.R.H, A.P., C.H., M.J.M, B.W.C., E.S., A.O., R.M., J.S., K.B., B.T.J, L.P., K.L., B.T., L.M.Y., Y.I.K., P.T.M. and F.L. performed research. W.S., L.R.H., C.H., E.S., J.W.B., P.T.M. and F.L. analyzed research; B.T., A.B., A.E.M, J.L.K, R.J.C., J.W.B. contributed clinical metadata and isolates. W.S., M.J.M, P.T.M and F.L. wrote the paper with input from all authors.

Ethics approval and consent to participate

The isolates and clinical information were collected as part of the public health surveillance activities of the MRSN, as determined by the WRAIR branch director and Human Subjects Protection Branch (HSPB) which granted ethical approval. Informed patient consent was waived as samples were taken under a hospital surveillance framework for routine sampling. The research conformed to the principles of the Helsinki Declaration.

Additional files

Supplemental Table 1. *Isolate Metadata.*

Supplemental Table 2. *All vs All SNP Comparison.*

Supplemental Table 3. *Environmental Isolates.*

Supplemental Table 4. *Antibiotic Susceptibility Testing.*

Supplemental Table 5. *Prediction of variants in outbreak isolates.*

Supplemental Table 6. *Resistome reference intrinsic alleles.*

Supplemental Table 7. *Mutations in intrinsic genes associated with antibiotic resistance.*

Supplemental Table 8. *Mutations in functional groups or pathways.*

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Reviewer #1 (Public review):

Summary:

This is a manuscript describing outbreaks of Pseudomonas aeruginosa ST 621 in a facility in the US using genomic data. The authors identified and analysed 254 P. aeruginosa ST 621 isolates collected from a facility from 2011 to 2020. The authors described the relatedness of the isolates across different locations, specimen types (sources), and sampling years. Two concurrently emerged subclones were identified from the 254 isolates. The authors predicted that the most recent common ancestor for the isolates can be dated back to approximately 1999 after the opening of the main building of the facility in 1996. Then the authors grouped the 254 isolates into two categories: 1) patient-to-patient; or 2) environment-to-patient using SNP thresholds and known epidemiological links. Finally, the authors described the changes of resistance gene profiles, virulence genes, cell wall biogenesis and signaling pathway genes of the isolates over the sampling years.

Strengths:

The major strength of this study is the utilisation of genomic data to comprehensively describe the characteristics of a long-term Pseudomonas aeruginosa ST 621 outbreak in a facility. This fills the data gap of a clone that could be clinically important but easily missed from microbiology data alone.

Weaknesses:

As the authors highlighted in the Discussion section, a limitation of this study is that there is potential sampling bias due to partial sampling of clinical P. aeruginosa isolates. However, the work is still important to showcase the potential benefits of applying genomic sequencing techniques to support infection prevention controls in hospital settings. The limitation on potential sampling bias could inspire further work to explore an optimal clinical isolate sampling framework for genomic analyses to support outbreak investigation. The other limitation that the authors have highlighted in the Discussion session is the lack of epidemiology data to support the interpretation of the inferred patient-to-patient and environment-to-patient transmissions, which emphasised the importance of metadata to complement genomic data analysis in outbreak investigation for future studies.

Impact of the work:

First, the work adds to the growing evidence implicating sinks as long-term reservoirs for important MDR pathogens, with direct infection control implications. Moreover, the work could potentially motivate investments in generating and integrating genomic data into routine surveillance. The comprehensive descriptions of the Pseudomonas aeruginosa ST 621 clones outbreak is a great example to demonstrate how genomic data can provide additional information about long-term outbreaks that otherwise could not be detected using microbiology data alone. Moreover, identifying the changes in resistance genes and virulence genes over time would not be possible without genomic data. Finally, this work provided additional evidence for the existence of long-term persistence of Pseudomonas aeruginosa ST 621 clones, which likely occur in other similar settings.

Comments on revisions:

The paper would be further strengthened from an additional timeline indicating when routine surveillance was introduced and examples of actions or changes guided by the surveillance data that resulted in decrease in ST 621 transmission. This additional information would be useful to support the final statement in the Abstract suggesting "Since initial identification, extensive infection control efforts guided by routine, near real- time surveillance have proved successful at slowing transmission."

https://doi.org/10.7554/eLife.93181.2.sa3

Reviewer #2 (Public review):

Summary:

The authors present a report of a large Pseudomonas aeruginosa hospital outbreak affecting more than 80 patients with first sampling dates in 2011 that stretched over more than 10 years and was only identified through genomic surveillance in 2020. The outbreak strain was assigned to the sequence type 621, an ST that has been associated with carpabapenem resistance across the globe. Ongoing transmission coincided with both increasing resistance without acquisition of carbapenemase genes as well as convergence of mutations towards a host-adapted lifestyle.

Strengths:

The convincing genomic analyses indicate spread throughout the hospital since the beginning of the century and provide important benchmark findings for future comparison

The sampling was based on all organisms sent to the Multidrug resistant Organism Repository and Surveillance Network across the U.S. Military Health System.

Using sequencing data from patient and environmental samples for phylogenetic and transmission analyses as well as determining recurring mutations in outbreak isolates allows for insights into the evolution of potentially harmful pathogens with the ultimate aim of reducing their spread in hospitals.

Weaknesses:

The epidemiological information was limited and the sampling methodology was inconsistent, thus complicating inference of exact transmission routes. Epidemiological data relevant for this analysis include information on the reason for sampling, patient admission and discharge data and underlying frequency of sampling and sampling results in relation to patient turnover.

Comments on revisions:

Thank you for the careful revision and consideration of my comments.

I am pleased to confirm that all my concerns have been comprehensively addressed.

The changes and additions made have resolved my initial feedback, and I see no need to alter my evaluation.

https://doi.org/10.7554/eLife.93181.2.sa2

Reviewer #3 (Public review):

Summary:

This paper by Stribling and colleagues sheds light on a decade-long P. aeruginosa outbreak of the high-risk lineage ST-621 in a US Military hospital. The origins of the outbreak date back to the late 90s and it was mainly caused by two distinct subclones SC1 and SC2. The data of this outbreak showed the emergence of antibiotic resistance to cephalosporin, carbapenems and colistin over time highlighting the emerging risk of extensively resistant infections due to P. aeruginosa and the need for ongoing surveillance.

Strengths:

This study, overall, is well constructed and clearly written. Since detailed information on floor plans of the building and transfers between facilities was available, the authors were able to show that these two subclones emerged in two separate buildings of the hospital. The authors support their conclusions with prospective environmental sampling in 2021 and 2022 and link the role of persistent environmental contamination to sustaining nosocomial transmission. Information on resistance genes in repeat isolates for the same patients allowed the authors to detect the emergence of resistance within patients. The conclusions have broader implications for infection control at other facilities. In particular, the paper highlights the value of real-time surveillance and environmental sampling in slowing nosocomial transmission of P. aeruginosa.

Weaknesses:

My major concern is that the authors used fixed thresholds and definitions to classify the origin of an infection. As such, they were not able to give uncertainty measures around transmission routes nor quantify the relative contribution of persistent environmental contamination vs patient-to-patient transmission. The latter would allow the authors to quantify the impact of certain interventions. In addition, these results represent a specific US military facility and the transmission patterns might be specific to that facility. The study also lacked any data on antibiotic use that could have been used to relate to and discuss the temporal trends of antimicrobial resistance.

Comments on revisions:

The authors have addressed my concerns adequately in the revised manuscript.

https://doi.org/10.7554/eLife.93181.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *This is a manuscript describing outbreaks of Pseudomonas aeruginosa ST 621 in a facility in the US using genomic data. The authors identified and analysed 254 P. aeruginosa ST 621 isolates collected from a facility from 2011 to 2020. The authors described the relatedness of the isolates across different locations, specimen types (sources), and sampling years. Two concurrently emerged subclones were identified from the 254 isolates. The authors predicted that the most recent common ancestor for the isolates can be dated back to approximately 1999 after the opening of the main building*

of the facility in 1996. Then the authors grouped the 254 isolates into two categories: 1) patient-to-patient; or 2) environment-to-patient using SNP thresholds and known epidemiological links. Finally, the authors described the changes in resistance gene profiles, virulence genes, cell wall biogenesis, and signaling pathway genes of the isolates over the sampling years.

Strengths:

The major strength of this study is the utilisation of genomic data to comprehensively describe the characteristics of a long-term Pseudomonas aeruginosa ST 621 outbreak in a facility. This fills the data gap of a clone that could be clinically important but easily missed from microbiology data alone.

Weaknesses:

The work would further benefit from a more detailed discussion on the limitations due to the lack of data on patient clinical information, ward movement, and swabs collected from healthcare workers to verify the transmission of Pseudomonas aeruginosa ST 621, including potential healthcare worker to patient transmission, patient-to-patient transmission, patient-to-environment transmission, and environment-to-patient transmission. For instance, the definition given in the manuscript for patient-to-patient transmission could not rule out the possibility of the existence of a shared contaminated environment. Equally, as patients were not routinely swabbed, unobserved carriers of Pseudomonas aeruginosa ST 621 could not be identified and the possibility of misclassifying the environment-to-patient transmissions could not be ruled out. Moreover, reporting of changes in rates of resistance to imipenem and cefepime could be improved by showing the exact p-values (perhaps with three decimal places) rather than dichotomising the value at 0.05. By doing so, readers could interpret the strength of the evidence of changes.

Impact of the work:

First, the work adds to the growing evidence implicating sinks as long-term reservoirs for important MDR pathogens, with direct infection control implications. Moreover, the work could potentially motivate investments in generating and integrating genomic data into routine surveillance. The comprehensive descriptions of the Pseudomonas aeruginosa ST 621 clones outbreak is a great example to demonstrate how genomic data can provide additional information about long-term outbreaks that otherwise could not be detected using microbiology data alone. Moreover, identifying the changes in resistance genes and virulence genes over time would not be possible without genomic data. Finally, this work provided additional evidence for the existence of long-term persistence of Pseudomonas aeruginosa ST 621 clones, which likely occur in other similar settings.

We thank the reviewer for their thorough evaluation of our work, and for the suggested improvements. A main goal of this study was to show that integrating routine wgs in the clinic was a game changer for infection control efforts. We appreciate this aspect was highlighted as a strength by this reviewer. While some of the weaknesses identified are inherent to the data (or lack thereof) available for this study, we have revised the manuscript to include a detailed discussion on limitations (sampling, thresholds of genetic relatedness, definition and categories etc.) that could influence the genomic inferences. We also provided exact p-values for the changes in rates of resistance, as requested. Finally, we have positively answered all the specific recommendations suggested by the reviewer and modified the manuscript accordingly.

Reviewer #2 (Public Review):

Summary:

The authors present a report of a large Pseudomonas aeruginosa hospital outbreak affecting more than 80 patients with first sampling dates in 2011 that stretched over more than 10 years and was only identified through genomic surveillance in 2020. The outbreak strain was assigned to the sequence type 621, an ST that has been associated with carpabapenem resistance across the globe. Ongoing transmission coincided with both increasing resistance without acquisition of carbapenemase genes as well as the convergence of mutations towards a host-adapted lifestyle.

Strengths:

The convincing genomic analyses indicate spread throughout the hospital since the beginning of the century and provide important benchmark findings for future comparison.

The sampling was based on all organisms sent to the Multidrug-resistant Organism Repository and Surveillance Network across the U.S. Military Health System.

Using sequencing data from patient and environmental samples for phylogenetic and transmission analyses as well as determining recurring mutations in outbreak isolates allows for insights into the evolution of potentially harmful pathogens with the ultimate aim of reducing their spread in hospitals.

Weaknesses:

The epidemiological information was limited and the sampling methodology was inconsistent, thus complicating the inference of exact transmission routes. Epidemiological data relevant to this analysis include information on the reason for sampling, patient admission and discharge data, and underlying frequency of sampling and sampling results in relation to patient turnover.

We thank the reviewer for their thoughtful feedback on our manuscript and for highlighting the quality of the genomic analyses. We agree that the lack of patient epi data (e.g. date of admission and discharge) and the inconsistent sampling through the years are limitations of this study. We have revised the manuscript to acknowledge these limitations and discuss how not having this data complicates the inference of exact transmission routes. Finally, we have positively answered all the specific recommendations suggested by the reviewer and modified the manuscript accordingly.

Reviewer #3 (Public Review):

Summary:

This paper by Stribling and colleagues sheds light on a decade-long P. aeruginosa outbreak of the high-risk lineage ST-621 in a US Military hospital. The origins of the outbreak date back to the late 90s and it was mainly caused by two distinct subclones SC1 and SC2. The data of this outbreak showed the emergence of antibiotic resistance to cephalosporin, carbapenems, and colistin over time highlighting the emerging risk of extensively resistant infections due to P. aeruginosa and the need for ongoing surveillance.

Strengths:

> *This study overall is well constructed and clearly written. Since detailed information on floor plans of the building and transfers between facilities was available, the authors were able to show that these two subclones emerged in two separate buildings of the hospital. The authors support their conclusions with prospective environmental sampling in 2021 and 2022 and link the role of persistent environmental contamination to sustaining nosocomial transmission. Information on resistance genes in repeat isolates for the same patients allowed the authors to detect the emergence of resistance within patients. The conclusions have broader implications for infection control at other facilities. In particular, the paper highlights the value of real-time surveillance and environmental sampling in slowing nosocomial transmission of P. aeruginosa.*
>
> *Weaknesses:*
>
> *My major concern is that the authors used fixed thresholds and definitions to classify the origin of an infection. As such, they were not able to give uncertainty measures around transmission routes nor quantify the relative contribution of persistent environmental contamination vs patient-to-patient transmission. The latter would allow the authors to quantify the impact of certain interventions. In addition, these results represent a specific US military facility and the transmission patterns might be specific to that facility. The study also lacked any data on antibiotic use that could have been used to relate to and discuss the temporal trends of antimicrobial resistance.*

We thank the reviewer for their evaluation of our work and for highlighting the broad implications of our findings regarding the application of real-time surveillance to suppress nosocomial transmission. We agree with the reviewer that fixed thresholds and definitions are imperfect to classify the origin of an infection. The design of this study (e.g. inconsistent sampling through time) was not conducive to provide a comprehensive/quantitative measurement of transmission routes. Thus, we decided to apply conservative thresholds of genetic relatedness and strict conditions (e.g. time between isolate collection, shared hospital location etc.) to favor specificity as our goal was simply to establish that cases of environmentto-patient transmission did happen. In the absence of a truth set, we have not performed sensitivity analysis, but we are conducting a follow-up study to compare inferences from MCMC models to our original fixed-thresholds predictions. This limitation is now discussed in the revised manuscript. Finally, we have positively answered all the specific recommendations suggested by the reviewer and modified the manuscript accordingly including the addition of Figure S3.

> ***Reviewer #1 (Recommendations For The Authors):***
>
> *The definitions used on lines 391-396 are necessarily somewhat arbitrary, but it would be helpful to have a little bit more justification for the choices made, particularly for the definition of environmental involving the "3x the number of years they were separated". It seems a little hard to square this with the more relaxed 10 SNP cutoff for a patient-to-patient designation. Are there reasons for thinking SNP differences associated with environmental transmission should be smaller than for patient-to-patient, or is the aim here just to set the bar higher for assuming an environmental source? Because these definitions are quite arbitrary, there could also be some value in exploring the sensitivity of the results to these assumptions.*

Thank you. We agree with the reviewers that SNP thresholds, albeit necessarily, are arbitrary and that more discussion/justification was needed to put the genomic inferences in context. We have revised the manuscript to indicate that: 1/ the 10 SNP cutoff for a patient-to-patient designation was set to account for the known evolution rate of P. aeruginosa (inferred by BEAST at 2.987E-7 subs/site/year in this study and similar to previous estimates PMID:

24039595) and the observed within host variability (now displayed in revised Fig. 1E). We note that this SNP distance was not sufficient and that an epi link (patients on the same ward at the same time) needed to be established. 2/ the environment-to-patient definition was indeed set to be most conservative (nearly identical isolates in two patients from the same ward with no known temporal overlap for > 365 days). This was indeed done to favor high specificity as this inference relied solely on clinical isolates (i.e. the identical environmental strain in the patientenvironment-patient chain was not sampled). For these clinical isolates to have acquired no/very little mutation in that much time, no/low replication is expected and, although unsampled, we propose this most likely happened on hospital surfaces.

> *While the term "core genome" should be familiar to most readers, "shell genome" and "cloud genome" are less widely known, and an explanation of what these terms mean here would be helpful.*

Thank you. We have revised the manuscript to define the core, shell, and cloud genomes as genes sets found in ≥ 99%, ≥ 95% and ≥ 15% of isolates, respectively.

> *In the first paragraph of the discussion, it could be added that in many cases for clinically important Gram negatives short read sequencing alone will fail to detect transmission events as outbreaks can be driven by plasmid spread with only very limited clonal spread (see, for example, https://www.nature.com/articles/s41564-021-00879-y)*

Thank you. We agree this is an important/emerging aspect of surveillance. However, the goal of this discussion point was to explain why such a large outbreak was missed prior to implementing WGS (short read) surveillance. We feel that discussing “plasmid outbreaks” (which is not at play here, and relatively rare in P. aeruginosa compared to the Enterobacteriaceae) and the need for long read will distract from the narrative.

> *line 599 What does "Mock" mean here? Would it be more accurate to say it is a simplified floor plan?*

Thank you. “Mock” was changed to “simplified”

> *IPAC abbreviation is only used once - spelling it out in full would increase readability.*

Revised manuscript was edited as suggested.

> *MHS is only used twice.*

Revised manuscript was edited to spell out Military Health System

> *Line 364: full stop missing.*

Revised manuscript was edited as suggested.

> *Line 401: Bayesian rather than bayesian.*

Revised manuscript was edited as suggested.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *Thank you for giving me the opportunity to review this interesting manuscript.*
>
> *The conclusions of this paper are mostly well supported by the data presented, but epidemiological information was limited and the sampling methodology was*

> *inconsistent, thus complicating inference of exact transmission routes.*
>
> *Major issues:*
>
> *What was the baseline frequency of clinical and/or screening samples of Pseudomonas aeruginosa at the hospital? Neither Figure 1D nor Table S1 allows for differentiating between clinical and screening samples. Most isolates were cultured from clinical materials, and there is no information about the patients' length of stay and their respective sampling dates. Is there any possibility of finding out whether the samples were collected for clinical or screening purposes? Would it be possible to include the patients' admission data to determine whether the strains were imported into the hospital or related to a previous stay, e.g. among known carriers? Also, the issue of sampling dates vs. patient stay on the ward should be addressed, as there may be an overlap in patients' stay on the ward but no overlap in terms of sampling dates or even missing samples (missing links).*

We have revised the manuscript to address this important point: i) 16 isolates were from surveillance swabs and are labelled "Surveillance" in Table S1. The remaining 237 were clinical isolates; ii) unfortunately, because the sampling was done under a public health surveillance framework, we do not have access to historical patient data (admission/discharge date, wards, rooms, etc.) and we can not calculate length of stay or better identify patient overlap. These limitations are now acknowledged in the discussion of the revised manuscript.

> *In order to evaluate the extent of the outbreak, more epidemiological data would be useful What is the size of the hospital, what is the average patient turnover, and what is the average length of stay in ICU and non-ICU? Is there any specialization besides the military label?*

We have revised the manuscript to indicate that facility A is 425-bed medical center and is the only Level 1 trauma center in the Military Health System. Unfortunately, the data to calculate length of stay, throughout the years, in ICU and non-ICU, was not available to us. This limitation is now also acknowledged in the discussion.

> *Perhaps the authors could attempwt to discuss the extent to which large outbreaks like these may be considered as part of unavoidable evolutionary processes within the hospital microbiome as opposed to accumulation and transmission of potentially harmful genes/clones, and differentiate between the putative community spread without any epidemiological links on the one hand, and hospital outbreaks that could be targeted by local infection prevention activities on the other hand.,*

We respectfully disagree with the suggestion that this large outbreak "may be considered as part of unavoidable evolutionary processes within the hospital microbiome" and should be opposed to "transmission of potentially harmful genes/clones". As a matter of fact, our data showed that infection control staff at Facility A responded with multiple interventions, including closing sinks, replacing tubing, and using foaming detergents. This resulted in slowing the spread of the ST621 outbreak with just 3 cases identified in 2022, 0 cases in 2023 and 1 case in 2024. This is now discussed in the revised manuscript.

> *Page 5, lines 88-92 lines 101-104. It seems as if the outbreak was identified only by the means of genomic surveillance. This raises questions as to the rationale for sampling and sequencing, especially prior to 2020. Considering 11 cases per year between 2011 and 2016, one could assume such an outbreak would have been noticed without sequencing data.*

The MRSN was created in 2010, in response to the outbreak of MDR Acinetobacter baumannii in US military personnel returning from Iraq and Afghanistan. Between 2011 and 2017, the MRSN collected MDR isolates (mandate for all MDR ESKAPE but compliance varied between years and facilities) from across the Military Health System and, for select isolates (e.g. high-risk isolates carrying ESBLs or carbapenemases) performed molecular typing by PFGE. In 2017 the MRSN started to perform whole genome sequencing of its entire repository. In 2020, a routine prospective sequencing service was started and first detected the ST621 outbreak. A retrospective analysis of historical isolate genomes (2011-2019) identified additional cases. The first paragraph of the discussion lists possible factors to explain why the ST621 escaped detection by traditional approaches. We believe 11 cases per year is not a strong signal when stratified by month, wards, or both, especially for a clone lacking a carbapenemase and without a remarkable antibiotic susceptibility profile.

> *Did the infection control personnel suspect transmission? If yes, was the sampling and submission of samples to the MRSN adapted based on the epidemiologic findings?*

The ST621 outbreak was unsuspected before the initial genomic detection in 2020. Until that point, MDR isolates only (Magiorakos et al PMID: 21793988) were collected but compliance was variable through time. Quickly thereafter (starting in 2021), complete sampling of all clinical P. aeruginosa (MDR or not) from Facility A was started. The manuscript was revised to clarify those details of the sampling strategy.

> *Is there any information about how many environmental sites were sampled without evidence of ST621 / screening samples were cultured without evidence of Pseudomonas aeruginosa?*

For patient isolates, only 16 isolates were from surveillance swabs. The remaining 237 were clinical isolates. No denominator data was available to calculate P. aeruginosa and ST-621 positivity rate in surveillance swabs throughout the time period. For environmental isolates, a total of 159 swabs were taken from 55 distinct locations in 8 wards/units including the ER. This data is now included in the revised manuscript. However, a complete analysis of these swabs (positivity rate for ESKAPE pathogens, P. aeruginosa, per ward/floor/room, per swab type (sink drain, bed rail etc.) etc.) is beyond the scope of this study and is being performed as a follow up investigation.

> *Page 5 lines 89 and 39 Figure S1B. Please describe how the allelic distance for the cluster threshold was selected.*

As indicated in the legend of Figure S1B, no thresholds were applied. All ST621 isolates ever sequenced by the MRSN were included. All except 3 isolates shared between 023 cgMLST allelic differences. The remaining 3 were distant by 88-89 allelic differences. The text was revised to clarify this point.

> *Page 5 lines 99-100. Could the authors please provide some distribution measures (e.g. IQR).*

Done as requested. The revised manuscript now reads "…of just 38 single nucleotide polymorphisms (SNPs), and an IQR of 19 (Fig. 1A, Table S1)."

> *Page 5 line 102. Could the authors please provide some distribution measures (e.g. IQR).*

Please see above. A chart was created and is now included as Fig. S2.

> *Page 6 line 107 and page 34 figure 1c. In the text it is stated that isolates were collected in 27 wards, the figure 1C depicts 26 wards and n/a.*

Thank you for spotting this inconsistency. This has been fixed in the revised manuscript.

> *Page 6 lines 117-118. Samples collected in the emergency room would imply samples collected on admission, already addressed previously. Did the authors investigate a potential import into the hospital from community reservoirs or were all these isolates collected among patients who had been previously admitted to the hospital and/or tested positive for the outbreak strain?*

We agree that samples collected in the ER imply samples collected on admission. Of the 29 ER isolates only 9 (31%) were primary isolates (first detection in a new patient) which suggests a majority were from returning patients at Facility A. Because the sampling was done under a public health surveillance framework, we do not have access to historical patient data (admission/discharge date, wards, rooms, etc.) to investigate/confirm that these 9 patients had previous visits at Facility A. This point is now discussed in the revised manuscript.

> *Page 6 line 128. This could also represent increased selective pressure. However, according to Table S1, the 28 isolates collected in 2011 (the number does not match with Figure 1D) were from many different wards, thus indicating earlier spread throughout the hospital.*

Yes, we agree. Please note that table S1 lists all isolates for 2011 whereas Figure 1D focuses on primary (first isolate from each patients) only.

> *Page 7 line 133. Both Figure 2 and the discussion section, page 13 line 296 suggest the year 2005 instead of 2004?*

Thank you for catching this typographical error. This was corrected to 2004 in the revised manuscript.

> *Figure 1E. The figure should also depict intra-patient diversity for comparison.*

Thank you for this great suggestion. We have revised Figure 1E accordingly.

> *Page 7, lines 146-147 Could the authors attempt explaining the upper part of the bimodal peaks?*

This is an all-vs-all SNP analysis for all inter-patient isolates. For each isolates all distances to other isolates are reported, not only the smallest. The upper peaks represent comparisons to isolates from a different outbreak subclone (SC1 vs SC2).

> *Page 7, line 150 This is a very small number considering the extent of the outbreak and suggests a large number of missing links. Or does this rather imply continuous import and evolution over time that does not necessarily represent transmission within the hospital?*

We believe all cases were due to transmission happening within the hospital. Based on conservative thresholds (genetic relatedness and epi link, or lack thereof) the precise origin from another patient (n=10) or a contaminated surface (n=12) can be inferred. For the remaining 60 patients, with the available sampling, the conditions we chose are not met and we simply do not conclude whether a direct patient-to-patient or an environmental origin was more likely.

> *Page 8 line 155. What does the temporal overlap refer to - sampling date versus patient's stay on the ward? Please specify.*

The temporal overlap was investigated from sampling dates, as dates of patient admission/discharged were not available.

> *Page 8, line 157: What does primary/serial isolate mean - first and follow-up samples of ST621 per patient?*

Yes. Primary isolate is used to designate the first isolate from a patient. Serial isolates designate follow-up samples of ST621.

> *Page 8 line 165: Table S3 and Figure 3 only refer to environmental samples from three wards. Ward 20 rooms 2 and 18 as well as ward 1 rooms 1 and 6 were hotspots - is there any information on the specific infection control/disinfection measures? Addressed in discussion page 12, lines 273-275, but no information on what was actually done.*

The manuscript was revised to indicate the precise disinfection measures that were taken. A follow-up study is ongoing to assess long-term efficacy and monitor possible retrograde growth from previously contaminated sinks.

> *Page 8 line 175: Evaluation of change in resistance fraction over time - There may have been a selection bias with an inconsistent number of strains sequenced per year.*

Yes, incomplete sampling and possible selection bias are now listed with other limitations of this study in the discussion of the revised manuscript.

> *Page 9 line 183: The referral to Table S1 is unclear, I could not find the number and the specific isolates selected for long-read sequencing.*

Thank you. This has been added to the revised Table S1.

> *Page 10 lines 217-225 and Figure 4C: Perhaps it is possible to better align what is written in the text and the caption of the figure. The caption does not clarify that only one patient develops colistin resistance (what was the reason to include the other patients?).*

Thank you. We have revised the text and the caption of the figure to clarify that only isolates from one patient developed colistin resistance. The isolates from the other patients on Fig. 4C are shown to provide context and accurately map the emergence of the PhoQE77fs mutation.

> *Page 10, lines 228-229 and Table S5: How is it possible to identify those 64 genes in Table S5?*

We have revised Table S5 to facilitate the identification of the 64 genes with ≥ 2 independently acquired mutations (excluding SYN). Specifically, we have added column E labeled “Counts independent mutations per locus (excluding SYN)”. A total of 205 rows (in this table each row is a variant) have a value ≥ 2 and these represent 64 genes (upon deduplication of locus tags).

> *Page 13, lines 280-281: Where is the information on chronic infection presented? Serial cultures would not necessarily mean chronic infection.*

Authors response: Yes, we agree this was not the appropriate characterization and this was revised to ‘long-term’ infections.

> *Page 14 line 306: Emergence of colistin resistance in a single patient, correct?*

Yes. This was further clarified in the text.

> *Page 14 lines 315-320: This should go to the results section. In particular disinfection, closing, and replacing of tubing should be mentioned in the results section in reference to the results presented in Table S3.*

Thank you. We have considered this suggestion and have decided to leave this discussion as the closing paragraph of this publication. A follow-up study is ongoing to assess long-term efficacy of these interventions on the ST-621 bur also other outbreak clones at Facility A.

> *Methods*
>
> *Page 15 lines 330-333: Perhaps it is possible to avoid redundancy.*

Thank you. We have revised the text accordingly.

> *Page 15 lines 341: Information on which isolates were subjected to long-read sequencing is missing.*

Thank you. This has been added to the revised Table S1.

> *Page 16 line 345: Was there a particular reason why Newbler was chosen?*

No. At the time Newbler was the default assembler built in the MRSN bacterial genome analysis pipeline and QC processes.

> *Page 16, line 357-358: What was the rationale for selecting this isolate as reference genome?*

This isolate was chosen because it was collected early in the outbreak and phylogenetic analysis revealed it had low root to tip divergence.

> *Page 16 line 361: Why 310 isolates, if only 253 were assigned to the outbreak clone and only a subset of those were collected in facility A?*

This was a typographical error that has corrected (it now reads “…set of 253 isolates.”) in the revised manuscript.

> *Page 17 lines 387-395: What is the reason that intra-patient diversity was not included in the set of criteria for SNP distances?*

The observed within host variability (now displayed in revised Fig. 1E) was taken into consideration when setting SNP thresholds for categorizing patient-to-patient transmission or environment-to-patient event. This is now clarified in the revised manuscript.

> *Page 17 line 392: How was the threshold of <=10 SNPs determined?*

The 10 SNP cutoff to infer a patient-to-patient transmission event was set to account for the known evolution rate of P. aeruginosa (inferred by BEAST at 2.987E-7 subs/site/year in this study, and similar to previous estimates PMID: 24039595) and the observed within host variability (now displayed in revised Fig. 1E). We note that this SNP distance was not sufficient and that an epi link (patients on the same ward within the same month) needed to be established.

> *Page 17 line 395 and Figure 2: What was the assumed average mutation rate per genome per year?*

Thank you. The mean substitution rate inferred by BEAST was 2.987E-7 similar to estimate from previous studies on P. aeruginosa outbreaks (e.g. PMID: 24039595).

> ***Reviewer #3 (Recommendations For The Authors):***
>
> *Please find (line-by-line comments) on each section of the manuscript below:*
>
> *Introduction*
>
> *Line 86: I am wondering why the authors state ">28 facilities" instead of the exact number of facilities from which these lineages were recovered.*

Thank you. Manuscript was revised to provide the exact number of facilities. It now reads "… recovered from 37 and 28 facilities, respectively."

> *Methods*
>
> *It's not clear to me which criteria were used for collecting these isolates (both prospective and retrospective). I understand that some of the data are described in more detail in Lebreton et al but I did not find the specific criteria for the collection of the isolates and I imagine that these might differ if different facilities. Would it be possible to comment on that and add a short paragraph in the Methods section?*

Thank you. This lack of clarity was also raised by other reviewers, and we have revised the manuscript to indicate that: 1/MDR isolates only (Magiorakos et al PMID: 21793988) were collected from 2011-2020 with the same criteria for all facilities although compliance was variable through time and between facilities; and 2/ starting in 2021 all P. aeruginosa isolates, irrespective of their susceptibility profile, were collected from Facility A

> *The data comes from a US Military hospital. Is this related to the US Veterans Affairs Healthcare system? Is there more detailed information about the demographics of the patient population?*

Facility A is part of the Military Health System (MHS) which provides care for active service members and their families. This is distinct from the US Veterans Affairs Healthcare system. Only limited patient data was accessible to us as this study was done as part of our public health surveillance activities. Patient age (avg. 57.2 +/- 21.0) and gender (ratio male/female 1.7) are provided in the revised manuscript.

> *Line 384ff: The origin of infection was inferred based on the SNP threshold and epidemiological links. However, recombination events can complicate the interpretation of SNP data. Have the authors attempted to account for this?*

Thank you. We agree that recombination events can complicate the interpretation of SNP data. We used Gubbins v2.3.1 to filter out recombination from the core SNP alignment, as indicated in the revised manuscript.

> *The authors' definition of environment-to-patient transmission seems conservative (nearly identical strain and no known temporal overlap for > 365 days). Have the authors changed the threshold, performed sensitivity analyses, and tested how this would affect their results?*

Indeed, acknowledging that fixed thresholds have limitations in their ability to accurately predict the origin of infections, we took a conservative approach to favor specificity as our goal was simply to establish that cases of environment-to-patient transmission did happen. In the absence of a truth set, we have not performed sensitivity analysis, but we are conducting a follow-up study to compare inferences from MCMC models to our original predictions. This limitation is now discussed in the revised manuscript.

> *The authors don't seem to incorporate the role of healthcare workers in the transmission process. Could they comment on this? I am assuming that environment-to-patient transmission could either be directly from the environment to the patient or via a healthcare worker. I think it's fine to make simplifying assumptions here but it would be great if this was explicitly described.*

Thank you for this suggestion. We have not sampled the hands of healthcare workers in this study. As a result, the reviewer is correct to say that we made the simplifying assumption that healthcare workers would be possible intermediates in either environment-topatient or patient-to-patient transmissions, as previously described by others (PMID: 8452949). This limitation is now discussed in the revised manuscript.

> *Page 5, line 100: What does "all vs all" mean? Based on the supplement, I assume it's the pairwise distance and then averaged across all of those. It would improve the readability of the manuscript if the authors could briefly define this term and then maybe refer to Table S1.*

Thank you. We have created Fig.S2 and revised the manuscript to state that ST-621 isolates from facility A belonged to the same outbreak clone with a distance (averaged all vs all pairwise comparison) of just 38 single nucleotide polymorphisms (SNPs), and an IQR of 19 (Fig. S2, Table S1).

> *Figure 1D: It would be interesting to see additional figures in the supplement on the percentage of sequenced isolates per year and whether it varies across the different sources/sites. Is there any information on which isolates were chosen for sequencing?*

Lack of clarity in the sampling/sequencing scheme was raised by multiple reviewers and we have provided a thorough response to earlier comments. We also have revised the material and methods section accordingly. Finally, we have created Fig. S3 to show the percentage of sequenced isolates per year across different sources/sites, as suggested by the reviewer. No noticeable patterns were observed.

> *It seems like only a subset of all clinical isolates were sequenced. Would it be possible that SC2 was present already earlier but not picked up until a certain date?*

Although all isolates received by the MRSN were sequenced, compliance varied through time so it is true that not all clinical isolates were sequenced between 2011-2019. As such, we fully agree with this hypothesis and discuss this possibility as BEAST analysis placed the origin of SC2 in 2004 while the first detection of an SC2 isolate was in December 2012. This limitation is now discussed in the revised manuscript.

> *Could the authors elaborate on whether the isolates resulted from single-colony picks? Is it possible that the different absence of a subclone is due to the fact that they picked only a colony?*

Yes, the isolates resulted from single-colony picks except when the presence of different colony morphologies was noted. In the latter, representative isolates for each colony morphologies were processed. We have revised the methods to make that clear.

> *Figure 2: It is difficult to see which nodes belong to which patient due to the small font size. I wonder if it was possible to color the nodes for each patient, to make it more readable.*

We tried coloring the nodes but with > 60 distinct patients/colors we decided it did not improve clarity. We have revised figure 2 to increase the font size.

> *Page 7-8, lines 154-155: Did the authors check whether there were isolates of the same strain (that were found in the environment) present in other patients elsewhere in the ward?*

Yes. In rare cases, we observed virtually genetically identical isolates from two patients collected in different wards. Because we only have access to clinical isolate data (collected from patient X in ward Y) and do not have access to patient data (admission/discharge date, wards, rooms, etc.), we do not know but cannot exclude that patients overlap in a room prior to the sampling of their P. aeruginosa isolates. We designed our fixed thresholds to be conservative. As a result, in this analysis, these cases are labelled as “undetermined”.

> *Page 8: Do the authors have any information on antibiotic use during this timeframe? From the discussion, it seems like there is no patient-level prescription data. Is there any data on overall trends? How were trends in antibiotic use correlated with trends in antibiotic resistance?*

Unfortunately, patient-level prescription data (or any other data not linked to the bacterial specimens) was not accessible to us as this study was done as part of our public health surveillance activities.

> *To infer the origin of infection, the authors used a static method with fixed thresholds and definitions. This study does not provide any uncertainty with their estimates. Maybe the authors could add a sentence in the discussion section that MCMC methods to infer transmission trees incorporating WGS could provide these estimates. These methods have not been applied to PA a lot but two examples where MCMC methods have been used without WGS (though the definition of environmental contamination may differ between these studies and this study).*
>
> *https://doi.org/10.1186/s13756-022-01095-x*
>
> *https://doi.org/10.1371/journal.pcbi.1006697*

Thank you for this great suggestion. We have revised the manuscript to include a discussion on the limitations of fixed thresholds to infer transmission chains/origins, and to discuss existing alternatives including MCMC methods.

> *Line 322-323: This sentence is a bit vague since not all of these HAI are due to P. aeruginosa. I would suggest citing a number that is specific to PA.*

Thank you. While our paper shows a particular example of protracted P. aeruginosa outbreak, the roll-out of routine WGS surveillance in the clinic will help prevent hospital-associated drug-resistant infections for more than this species. We believe that broadening the scope in the last sentence of the manuscript is important and we decline to revise as suggested.

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